INCH-POUND

MIL-PRF-12126G
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SUPERSEDING
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20 February 1996

PERFORMANCE SPECIFICATION

VALVES, GATE: BRONZE, LEVER OPERATED, QUICK OPENING, 125 PSI AND 150 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers lever-operated, quick-opening, bronze gate valves with a solid-wedge or double-wedge disk and threaded ends.

1.2 Classification. The valves will be of the following classes, sizes, and wedge shape, as specified (see 6.2):

Class 1 - 125 pounds per square inch (psi) (862 kilopascals (kPa))
Class 2 - 150 psi (1034 kPa)

Sizes: 1-inch (25 millimeter (mm))
1.5-inch (38 mm)
2-inch (51 mm)

Wedge: Solid - A
Double - B

Comments, recommendations, additions, deletions, clarifications etc. and any data that may improve this document should be sent to U.S. Army Ground Vehicle System Center, ATTN: FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil/.

AMSC N/A FSC 4820

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	-	Unified Inch Screw Threads (UN and UNR Thread Form)

(Copies of this document are available online at www.asme.org).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org).

ASTM INTERNATIONAL

ASTM A123/A123M	-	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A197/A197M	-	Standard Specification for Cupola Malleable Iron
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B62	-	Standard Specification for Composition Bronze or Ounce Metal Castings

(Copies of these documents are available online at www.astm.org).

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SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE AS478	-	Identification Marking Methods
SAE J429	-	Mechanical and Material Requirements for Externally Threaded Fasteners

(Copies of these documents are available online at www.sae.org).

GENERAL MOTORS (GM)

GM9540P	-	Accelerated Corrosion Test - Do Not Use on New Programs; Replaced by GMW14872

(Copies of this document are available online at www.gm.com).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The valve shall be a quick-opening, lever-operated gate valve with a solid-wedge or a double-wedge disk as specified in the contract (see 6.2). The disk shall be actuated through an arrangement that shall provide a fast and positive valve action. The valve body with cap and valve stem shall be fabricated of bronze, and the internal working parts shall be brass. The hand lever shall be malleable iron. The valve end connections shall be integral with the valve body and shall be threaded internally with American Standard taper pipe threads (see 3.4). The exterior of the valve ends shall be shaped to provide wrench grip. The valve shall be constructed so that the stem packing may be renewed when the valve is closed and under pressure.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2. First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent valves delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.3.2 and 3.5).

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3.3.1 Material deterioration, prevention and control. The valves shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the valves may be exposed. Corrosion protection of materials other than ASTM B61 and ASTM B62 shall be equal to or exceed that provided by hot dip galvanized 1020 steel with coating thickness in accordance with ASTM A123/A123M with zinc phosphate pretreatment (see 4.4.2.3).

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with the component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the valves (see 6.5).

3.3.3 Bronze.

3.3.3.1 External parts. External parts and valve bodies, caps, and stems shall conform to ASTM B61 and shall meet all environmental and performance requirements of this document.

3.3.3.2 Internal parts. Internal working parts and valve disks shall be bronze conforming to ASTM B62 and shall meet all environmental and performance requirements of this document.

3.3.4 Handle levers. Hand levers shall be made from malleable iron conforming to ASTM A197/A197M.

3.3.5 Bolts. Bolts shall be made from steel conforming to SAE J429.

3.4 Threads. All threads shall conform to ASME B1.1.

3.5 Performance. The valve disk shall not jam and the internal working parts shall not be damaged by rapid and forceful operation of the lever. The valve body shall withstand the hydrostatic test pressure specified in table I without any evidence of leakage or seepage on the valve surface. When open, the valve shall provide unrestricted free-flow passage equal to that of the nominal pipe size of the valve. When closed, the valve shall withstand the full working pressure specified in table I applied from either end of the valve and shall show no evidence of leakage or seepage through or around the disk or seat (see 4.4.2).

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3.5.1 Pressure ratings. The valves shall be designed for the pressure ratings shown in table I.

TABLE I. Pressure ratings.

Class	Working pressure psi (kPa)	Hydrostatic test pressure at 150 °F (66 °C) psi (kPa)
1	125 (862)	250 (1724)
2	150 (1034)	300 (2068)

3.6 Valve caps. The valve shall be provided with a removable cap for access to operating linkage and valve disks. When the valve cap is of the threaded type, it shall be shaped to provide a wrench grip. Bolted-type caps shall not have less than four bolts of the capscrew type to secure the caps to the valve body.

3.7 Functional labeling. The valve shall be provided with double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (like “OPEN,” “CLOSE”) (see 3.9).

3.8 Treatment and painting. Parts normally painted shall use optional processes to treat and paint external surface which shall meet all performance and environmental requirements of this specification. Unless otherwise specified (see 6.2), the top coat color shall be in accordance with SAE AMS-STD-595, color No. 34079.

3.9 Identification marking. Valves shall be identified in accordance with SAE AS478.

3.10 Workmanship. The valves shall be free from defects such as corrosion, cracks, and other defects that could impair operation or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. The valve shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

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4.2.2 Tests. The valve shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Lot. All valves of the same class and size offered for inspection at one time shall constitute a lot.

4.3.2 Sampling. Sampling for examination and tests shall conform to ASQ Z1.4.

4.3.3 Examination. Samples selected as specified in 4.3.2 shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.4 Tests. Samples selected as specified in 4.3.2 shall be tested as specified in 4.4.2, except that the test pressure for all tests shall be held for a period of 1 minute. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The valve shall be examined as specified in table II.

TABLE II. Examination schedule.

Category	Defect	Method of examination
Major:		
101	Valve stem packing cannot be renewed when valve is closed and under pressure (see 3.1).	Visual
102	Wrench grip not provided on end connections (see 3.1).	Visual
103	Materials not as specified (see 3.3).	Visual
104	Materials not resistant to corrosion and deterioration, or treated to resist corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).	SIE 1/
105	Dissimilar metals are not effectively insulated from each other (see 3.3.1.1).	Visual
106	Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).	Visual
107	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the valves (see 3.3.2).	Visual
108	Threads not as specified (see 3.4).	SIE

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TABLE II. Examination schedule - Continued.

Category	Defect	Method of examination
109	Valve disk jammed or internal working parts damaged under rapid and forceful operation of the lever (see 3.5).	Visual
110	Flow passage not equal to that of the nominal pipe size (see 3.5).	SIE
111	Valve pressure ratings not as specified (see 3.5.1).	SIE
112	Valve caps not as specified (see 3.6).	SIE
113	Functional labeling not as specified (see 3.7).	Visual
114	Treatment and painting not as specified (see 3.8).	Visual
115	Color not as specified (see 3.8).	Visual
116	Identification marking missing, incomplete, incorrect, or illegible (see 3.9).	Visual
117	Workmanship not as specified (see 3.10).	Visual/SIE

1/SIE = Standard Inspection Equipment.

4.4.2 Tests.

4.4.2.1 Hydrostatic test, body. The valve body shall be subjected to the applicable hydrostatic test pressure and temperature specified in table I. Water shall be used as the test fluid. With one end of the valve closed and the gate in open position, pressure shall be applied from the other end. The test pressure shall be held for not less than 15 minutes. The valve body shall be examined for leakage or seepage. Any evidence of leakage or seepage shall constitute failure of this test (see 3.5 and 4.2.2).

4.4.2.2 Hydrostatic test, disk. The valve disk shall be subjected to the applicable working pressure specified in table I with the pressure applied at one end of the valve with the disk closed. Water shall be used as the test fluid. The pressure shall then be relieved, the valve shall be opened and closed, and the same pressure shall be applied at the other end. The test pressure shall be held for not less than 15 minutes. The valve shall be examined for leakage or seepage through or around the disk or seat. Any evidence of such leakage or seepage shall constitute failure of this test (see 3.5 and 4.2.2).

4.4.2.3 Corrosion protection. Corrosion protection shall meet the requirements of the Accelerated Corrosion Test GM9540P, Method B, until one of the items fail with defects such as extensive corrosion at scribe or significant penetration of base metal (see 3.3.1).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging

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activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves covered by this specification are for general purpose use with water, oil and liquid gas.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Class, size and part identifying number of valve required (see 1.2 and 6.6).
c. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.3).
d. Whether solid-wedge or double-wedge disk valve is required (see 3.1).
e. When first article is required (see 3.2).
f. Color required if other than as specified (see 3.8).
g. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

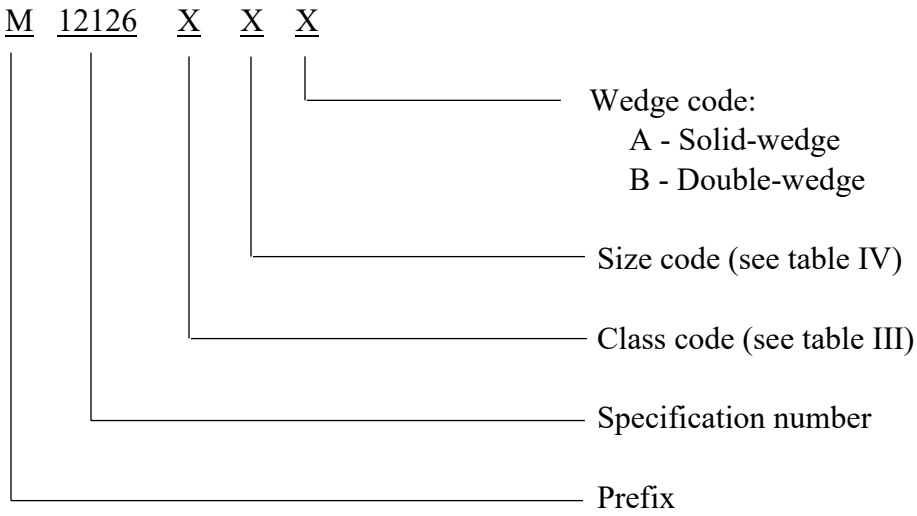
6.4 Subject term (key word) listing.

Double-wedge
Liquid gas
Oil
Solid-wedge
Threaded ends
Water

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6.5 Recovered materials. Recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.3.2).

6.6 Part or Identifying Number (PIN). The part or identifying number (PIN) of valves covered by this specification are designated in the following form (see 6.2):



Example: M12126A2B is the part number for a lever operated bronze gate valve, (A) class 1 (125 psi), (2) size 1.5 inch, (B) double wedge conforming to MIL-PRF-12126.

TABLE III. Class code.

Code	Class	Description
A	1	125 psi
B	2	150 psi

TABLE IV. Size code.

Code	Size	Description
1	1	1-inch
2	1.5	1.5-inch
3	2	2-inch

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - SH

Review activities:
Army - CE
Navy - YD
DLA - CC

Preparing Activity:
Army - AT

(Project 4820-2021-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at https://assist.dla.mil.

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